Rechteck

$$f_0(t) = \begin{cases} h & 0 \le t < \frac{L}{2} \\ -h & \frac{L}{2} < t < L \end{cases}$$

$$a_n = \frac{2}{L}\int_0^L f(t)\cos\left(\frac{2\pi n}{L}t\right)dt \qquad b_n = \frac{2}{L}\int_0^L f(t)\sin\left(\frac{2\pi n}{L}\cdot t\right)$$

$$a_0 = \frac{2}{L}\int_0^L f(t)\,dt = \frac{2h}{L}\left(\int_0^{L/2} 1\,dt - \int_{L/2}^{L} 1\,dt\right) = 0$$

$$a_n = \frac{2}{L}\int_0^{L/2}\cos\left(\frac{2\pi n}{L}\cdot t\right)dt - \int_{L/2}^{L}\cos\left(\frac{2\pi n}{L}\cdot t\right)dt = \frac{2h}{L}\left[\frac{\sin\left(\frac{2\pi n}{L}\cdot t\right)}{\frac{2\pi n}{L}}\right]_0^{L/2} + [\ldots \sin(\ldots)]$$

$$= 0$$

$$b_n = \frac{2h}{L}\left(\int_0^{L/2}\sin\left(\frac{2\pi n}{L}\cdot t\right)dt - \int_{L/2}^{L}\sin\left(\frac{2\pi n}{L}\cdot t\right)dt\right)$$

$$= \frac{2h}{L}\left(\left[\frac{-\cos\left(\frac{2\pi n}{L}t\right)}{\frac{2\pi n}{L}}\right]_0^{L/2} + \left[\frac{\cos\left(\frac{2\pi n}{L}\cdot t\right)}{\frac{2\pi n}{L}}\right]_{L/2}^{L}\right)$$

$$= \frac{h}{\pi n}\left((1-(-1)^n) + 1 - (-1)^n\right)$$

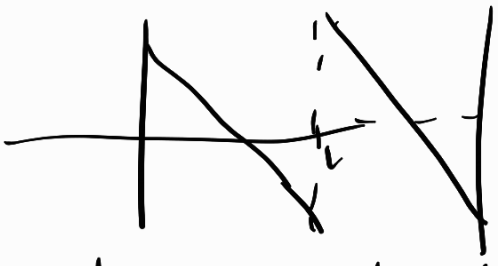
$$= \frac{4h}{\pi(2n+1)} \qquad n \in \mathbb{N}$$

$$f(t) = \frac{4h}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\sin\left(\frac{2\pi(2n-1)}{L}\cdot t\right) \qquad \frac{2\pi}{L} = w$$

$$= \frac{4h}{\pi}\left(\sin(wt) + \frac{1}{3}\sin(3wt) + \frac{1}{5}\sin(5wt) \ldots\right.$$

$$1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13}, \frac{1}{15}, \frac{1}{17}$$

Sägezahn



$$f(t) = h - \frac{2h}{L}t$$
$$= h\left(1 - \frac{2t}{L}\right)$$

punktsymmetrisch zur Intervallmitte sinusterme

Es $a_0 = 0$ weil im Mittel 0 $\int_0 f(t) = 0$

$$b_n = \frac{2}{L}\int_0^L \left(h - \frac{2h}{L}\cdot t\right)\sin\left(\frac{2\pi n}{L}\cdot t\right)dt$$

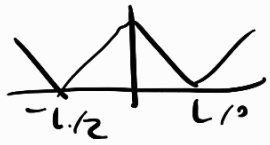
$$= \frac{2h}{L}\left(\left[\frac{\cos\left(\frac{2\pi n}{L}t\right)}{\frac{2\pi n}{L}}\right]_0^L + \left[\frac{2}{L}t\,\frac{\cos\left(\frac{2\pi n}{L}\cdot t\right)}{\frac{2\pi n}{L}}\right]_0^L - \int_0^L \frac{2}{L}\,\frac{\cos(\ldots)}{\frac{2\pi n}{L}}\,dt\right.$$

$$= \frac{h}{\pi n}\left(1 - 1 + 2 - \frac{2h}{L}\left[\frac{\sin\left(\frac{2\pi n}{L}\cdot t\right)}{\frac{2\pi n}{L}}\right]_0^L\right.$$

$$= \frac{2h}{\pi n} \qquad f(t) = \frac{2h}{\pi}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{2\pi n}{L}\cdot t\right)}{n}$$

$$f(t) = \frac{2h}{\pi}\left(\sin(wt) + \frac{1}{2}\sin(wt) + \frac{1}{3}\sin(wt) \ldots\right. \quad 1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13}, \frac{1}{15}, \frac{1}{17}, \frac{1}{19}$$

Fourierkoeffizienten Dreieck



$$f_0(t) = \begin{cases} \left(t + \frac{L}{2}\right)\frac{2}{L}\cdot h & -\frac{L}{2} < t < 0 \\ \left(\frac{L}{2} - t\right)\frac{2}{L}\cdot h & 0 \le t < \frac{L}{2} \end{cases}$$

Achsensymmetrie um 0 nur Cosinusterme a_n

$$a_0 = \frac{2}{L}\int_{-\frac{L}{2}}^{L/2} f_0(t)\,dt$$

$$= \frac{4}{L}\int_0^{L/2} h - \frac{2t\cdot h}{L}\,dt = \frac{4}{L}\left[ht - \frac{t^2 h}{L}\right]_0^{L/2}$$

$$= 2h - h = h \qquad w = \frac{2\pi}{L}$$

$$a_n = \frac{2}{L}\int_{-L/2}^{L/2} f_0(t)\cos(nwt)\,dt = \frac{4}{L}\int_0^{L/2}\left(h - \frac{2t\cdot h}{L}\right)\cos(wnt)\,dt$$

$$= \frac{4}{L}\left(\left[\left(h - \frac{2t\cdot h}{L}\right)\frac{\sin(wnt)}{nw}\right]_0^{L/2} + \int_0^{L/2}\frac{2h}{L}\,\frac{\sin(wnt)}{nw}\,dt\right.$$

$$= \frac{4}{L}\left[\frac{-2h}{L}\,\frac{\cos(wnt)}{w^2}\right]^{L/2} = \frac{4}{Lw^2}\,\frac{2h}{L}\left(1 - (-1)^n\right)$$

$$= \frac{4h}{n^2\pi^2} \qquad n\text{-ungerad} \qquad \frac{4h}{(2n+1)^2\pi^2}$$

$$f(t) = \frac{h}{2} + \frac{4h}{\pi^2} \sum \frac{1}{(2n-1)^2} \cos(w(2n-1)t)$$

$$= \frac{h}{2} + \frac{4h}{\pi^2}\left(\cos(wt) + \frac{1}{9}\cos(3wt) + \frac{1}{25}\cos(5wt)\right) \ldots$$

$$1, \frac{1}{9}, \frac{1}{25}, \frac{1}{49}, \frac{1}{81}, \frac{1}{121}, \frac{1}{169}, \frac{1}{225}, \frac{1}{289}$$